



Reflexive AI

REFLEXIVE AI RESEARCH INITIATIVE

Human-Centered AI

Overview of relevant insights from scientific research

10 minutes

details & sources @ <https://github.com/LucidInstitute/Reflexive-AI-Initiative/tree/main/Literature>

DEFINING AI

AI can take over human thinking, and technology is never neutral.

// KEY FINDINGS

- 1 OECD definition.** A machine-based system that infers from input how to generate predictions, content, recommendations or decisions (OECD, 2024). More simply: “Computers that do the sorts of things minds can do” (Boden, 2016).
- 2 Minds, delegated.** Unlike earlier revolutions, thinking and creativity can now be delegated, not only physical action. (Boden, 2016; Hilbert, 2020). Robotics will also replace more physical action.
- 3 Technology embodies power.** It shapes how we see the world and imposes behaviors on people. (Winner, 1985; Latour, 1992)
- 4 Bias carries over.** AI can perpetuate biases already present in its training data. (Barocas & Selbst, 2016; Noble, 2018)

Individual gains are real, but they do not automatically become team gains.

// KEY FINDINGS

1

Firm level. Positive productivity effects across sectors, stronger when AI is integrated into work activities. (Necula et al., 2024; Czarnitzki et al., 2023)

2

Employee level. In a survey of 359 bank employees, AI use raised productivity and engagement. (Gusti et al., 2024)

3

Perception gap. Developers felt more productive, yet commit activity did not change and experienced developers were not faster. (Stray et al., 2026; Becker et al., 2025)

4

Lost downstream. Local gains are absorbed by workflow and organizational constraints. (DORA, 2025; Russo, 2026)

Whether AI helps or harms depends on trust, fit and culture.

// KEY FINDINGS

1

Engagement. AI raised engagement in two manufacturers, unless self-efficacy is low or jobs were cut. (Tortorella et al., 2025)

2

Trust. Low employer trust sharply reduces engagement; AI may bring an “alienational” contract. (Braganza et al., 2021)

3

Preconditions. Task fit, transparency and expertise. Black boxes breed aversion; bureaucracy builds “algorithmic cages”. (Bankins et al., 2023; Meijer et al., 2021)

4

Six displacements. Software engineers report epistemic, identity, normative, phenomenological, social and meaning shifts. (Tavantzis et al., in press)

AI can amplify existing inequality unless we design purposefully against it.

// KEY FINDINGS

- 1 Bias in practice.** A widely used health system under-recommended treatment for black relative to white patients. (Obermeyer et al., 2019)
- 2 Hard to remove.** Mitigation strategies exist, but implicit biases are difficult to eradicate. (Nazer et al., 2023; Howard & Borenstein, 2017)
- 3 Digital divide.** Inequality can grow between households that can use AI and those that cannot. (Cornelli et al., 2023)
- 4 Labor market.** Automation can replace low-skilled tasks, leaving some workers worse off despite seeming economic growth. (Agrawal et al., 2019; Li & Nichols, 2026)

Human-centered AI keeps people in control, backed by real governance – not just principles.

// KEY FINDINGS

1

Two dimensions. Human control and computer automation are separate axes, not a single scale. (Shneiderman, 2020)

2

Copenhagen Manifesto. 35 scholars set five values and twelve principles built on dignity, autonomy and wellbeing. (Russo et al., 2024)

3

Working definition. Efficient work while people stay in control, motivated, connected, and free to question or reject AI decisions. (Initiative draft)

4

Governance gap. Ethical guidelines say what, not how, and can lose to commercial interests. Structures and processes are needed. (Mittelstadt, 2019; Mäntymäki et al., 2022). It's not enough to say what is needed; we have to think *how* to do those things in practice.

THANK YOU

From insight to responsible practice